

	StudentID	Name	GradYear	CName	CGrade	Date_grades	Department
1							
2							
3	1	Linda Jones	2026	Data Visualization	3.9	2026-01-18	CS
4	1	Linda Jones	2026	Database Systems	2.3	2026-04-18	Math
5	2	Robert Smith	2028	AI Agents	3.8	2026-04-23	History
6	2	Robert Smith	2028	Linear Algebra	2.3	2026-05-16	Math
7	2	Robert Smith	2028	Quantum Physics	2.6	2026-05-12	CS
8	3	Mary Davis	2026	Database Systems	3.4	2026-01-24	Math
9	3	Mary Davis	2026	Linear Algebra	2.8	2026-01-21	Math
10	3	Mary Davis	2026	Machine Learning	3.9	2026-01-17	Math
11	4	Mary Jones	2026	AI Agents	2.9	2026-04-11	History
12	4	Mary Jones	2026	Data Visualization	2.7	2026-02-25	CS
13	5	Patricia Garcia	2026	Data Visualization	2.7	2026-01-28	CS
14	5	Patricia Garcia	2026	Database Systems	2.4	2026-04-24	Math
15	6	Mary Jones	2026	None	None	None	None
16	7	James Williams	2025	Database Systems	3	2026-04-23	Math
17	8	Robert Jones	2027	AI Agents	3.7	2026-04-12	History
18	8	Robert Jones	2027	Data Visualization	3.3	2026-01-10	CS
19	8	Robert Jones	2027	Machine Learning	3.7	2026-04-17	Math
20	9	Robert Davis	2027	Linear Algebra	2.3	2026-01-16	Math

--- Checking FD: CName -> Department ---

VIOLATION: Course 'Database Systems' associated with multiple departments: {'Math', 'CS'}

VIOLATION: Course 'Machine Learning' associated with multiple departments: {'Math', 'Art', 'CS'}

VIOLATION: Course 'Intro to Python' associated with multiple departments: {'Biology', 'CS', 'History'}

--- Checking FD: {CName, StudentID} -> CGrade ---

VIOLATION: Student 99 in 'Machine Learning' has multiple grades: {'0.0', '4'}

VIOLATION: Student 98 in 'Database Systems' has multiple grades: {'1.0', '2.5'}

SQL Detection: CName -> Department

Violation: Course 'Database Systems' is associated with departments: Math,CS

Violation: Course 'Intro to Python' is associated with departments: CS,History,Biology

Violation: Course 'Machine Learning' is associated with departments: Math,Art,CS

SQL Detection: {CName, StudentID} -> CGrade

Violation: Student 98 in course 'Database Systems' has grades: 1.0,2.5

Violation: Student 99 in course 'Machine Learning' has grades: 4,0.0

```
Original SQL Execution Time: 0.000209 seconds
[('2026-01-10', 4, 4.0, 2027), ('2026-01-11', 4, 3.5, 2025), ('2026-01-12', 3, 3.0, 2025), ('2026-01-13', 4, 4.0, 2025), ('2026-01-14', 4, 4.0, 2025), ('2026-01-15', 4, 3.3, 2025)]
View-Based SQL Execution Time: 0.000242 seconds
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Date          | Max Credits | Avg Credits | Min GradYear
-----
2026-01-10    | 4           | 4.00        | 2027
2026-01-11    | 4           | 3.50        | 2025
2026-01-12    | 3           | 3.00        | 2025
2026-01-13    | 4           | 4.00        | 2025
2026-01-14    | 4           | 4.00        | 2025
6.c Python TXT Execution Time: 0.000243 seconds
```